Multimodal Fusion of EHR and ECG Time-Series for Explainable Heart Failure Risk Stratification

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***Abstract*—Heart failure (HF) remains a leading cause of global mortality, requiring precise, long-term risk stratification for effective clinical management. Traditional machine learning models often rely on static Electronic Health Records (EHR) or analyze electrocardiogram (ECG) waveforms in isolation, failing to capture the complex, multimodal nature of cardiac exacerbation. In this paper, we propose a novel dual-branch deep learning architecture that simultaneously processes tabular clinical features and temporal ECG signals. Utilizing a Multi-Layer Perceptron (MLP) for 11-feature EHR data and a 1D-Convolutional Neural Network (1D-CNN) for 1000-step ECG waveforms, the model fuses the embeddings using a Gated Attention Mechanism to predict mortality risk (0.0 to 1.0). To bridge the gap between artificial intelligence and clinical utility, we introduce a deterministic post-processing Clinical Rules Engine. This engine maps the AI-derived risk probabilities and raw Ejection Fraction (EF) directly to established New York Heart Association (NYHA) functional classes and ACC/AHA treatment stages. Furthermore, SHapley Additive exPlanations (SHAP) are integrated to provide feature-level transparency, while a decoupled Express/Flask microservice architecture leverages a generative Large Language Model (LLM) to automatically draft physician-facing clinical narratives.**

***Index Terms*—Multimodal Fusion, Deep Learning, Heart Failure, Explainable AI, Clinical Decision Support.**

I. Introduction

Heart failure is a complex clinical syndrome characterized by the heart's inability to pump sufficient blood to meet the body's metabolic demands. Despite advancements in Guideline-Directed Medical Therapy (GDMT), predicting acute exacerbation events remains a significant clinical challenge.

Current predictive methodologies in clinical settings often rely on static risk scores that fail to account for the dynamic interplay between a patient's electrical cardiac rhythm and their systemic metabolic state. While artificial intelligence has shown promise, existing models predominantly suffer from unimodal constraints and the "Black Box" paradigm, where deep neural networks output raw probabilities without clinical context.

To solve this, the core objective function of our system shifts to a multimodal probability space:

$$P(\text{Exacerbation}) = f_\theta(X_{EHR}, X_{ECG}) \tag{1}$$

Where X_{EHR} represents the 1×11 dimensional tensor of clinical vitals, and X_{ECG} represents the $1 \times 1000 \times 1$ dimensional tensor of the raw heartbeat waveform. This paper introduces a unified pipeline that bridges these gaps through a dual-branch fusion architecture and a deterministic translation layer anchored to ACC/AHA guidelines.

II. Related Work

The application of machine learning for heart failure prediction has evolved significantly, though it remains largely fragmented across different data modalities.

A. EHR-Based Predictive Modeling

Historically, models predicting HF mortality have relied on tabular data. Algorithms such as Random Forests and XGBoost are frequently trained on datasets like MIMIC-IV to find nonlinear patterns in blood work [2]. Recent advancements have introduced architectures like Hi-BEHRT to track longitudinal clinical events [4], and broader multimodal frameworks for cardiovascular risk [3]. While computationally efficient for triage [6], models treating data statically miss crucial electrical precursors.

B. ECG-Based Deep Learning

Conversely, 1D-Convolutional Neural Networks (1D-CNN) have achieved cardiologist-level accuracy in detecting arrhythmias. However, an ECG cannot detect metabolic markers like kidney failure or hyponatremia. Models trained exclusively on ECG data lack the holistic physiological context necessary for long-term patient management.

C. Explainable AI in Healthcare

The adoption of complex fusion models in Clinical Decision Support Systems [5] is hindered by a lack of explainability [10]. The integration of SHAP (SHapley Additive exPlanations) [8] provides a mathematical breakdown of feature impact, a methodology proven effective in recent heart failure interpretability studies [7]. Furthermore, integrating these explainable outputs with generative transformer models [9] bridges the gap between raw data and actionable clinical guidelines [11], particularly for patients requiring advanced mechanical circulatory support [12].

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \tag{2}$$

Our proposed architecture builds upon these foundations by anchoring AI outputs directly to clinical guidelines via a deterministic rules engine.

III. Methodology

Our proposed Clinical Decision Support (CDS) system is structured into three primary computational phases: multimodal data preprocessing, dual-branch neural network inference, and deterministic clinical post-processing via a microservice architecture.

A. Data Acquisition and Preprocessing

The model utilizes retrospective cohorts from the MIMIC-IV (Medical Information Mart for Intensive Care) and MIMIC-IV-ECG databases, sourced from the Beth Israel Deaconess Medical Center. To ensure the model learned generalizable cardiovascular deterioration patterns, strict inclusion and exclusion criteria were applied.

1) Cohort Selection and Demographics: Patients were included if they had a documented primary or secondary ICD-10 diagnosis code associated with heart failure (e.g., I50.1, I50.2, I50.3) and possessed at least one corresponding 12-lead or Lead II continuous ECG recording within 24 hours of their laboratory draws. Patients under the age of 18 or those with severe missing data (defined as $> 30\%$ null values across the 11 target features) were excluded. The final synthesized cohort demographics are detailed in Table I.

TABLE I
Baseline Patient Demographics and Clinical Features

Feature	Value / Distribution	Unit
Total Patients	14,250	-
Age (Mean $\pm$ SD)	68.4 ± 12.1	Years
Gender (Male / Female)	58% / 42%	Percentage
Systolic BP (Mean)	124.5 ± 18.2	mmHg
Serum Creatinine (Mean)	1.4 ± 0.8	mg/dL
BUN (Mean)	28.5 ± 14.1	mg/dL
Sodium (Mean)	137.2 ± 4.5	mEq/L
LVEF% (Mean)	38.2 ± 11.4	Percentage

2) EHR Tabular Normalization: A subset of 11 critical clinical features was extracted from the EHR. Missing values were imputed using median statistical filling. Because neural networks are highly sensitive to unscaled variance (e.g., Sodium values range in the 130s while Creatinine ranges from 0.5 to 5.0), all continuous variables were normalized using Standard Scaling. For each feature x_i, the scaled value $x_{scaled}^{(i)}$ was computed as:

$$x_{scaled}^{(i)} = \frac{x_i - \mu}{\sigma} \tag{3}$$

Where μ is the mean of the training samples and σ is the standard deviation. This resulted in a standardized input tensor shape of $(1, 11)$.

3) ECG Time-Series Processing: Raw Lead II ECG signals were sampled at 500 Hz. To isolate the true cardiac electrical activity, baseline wander and high-frequency muscular artifacts were removed using a Butterworth bandpass filter (0.5 Hz to 50 Hz). The transfer function of the Butterworth filter is defined as:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}} \tag{4}$$

Where n is the filter order and ω_c is the cutoff frequency. Following filtration, the continuous waveforms were segmented into discrete, 2-second windows and Z-score normalized, resulting in an input tensor shape of $(1, 1000, 1)$.

B. Dual-Branch Neural Network Architecture

To capture both systemic metabolic degradation and immediate electrical arrhythmias, we designed a Y-shaped deep learning model utilizing Keras/TensorFlow.

1) Tabular Feature Extractor (Branch 1): The 11-dimensional EHR tensor is passed through a Multi-Layer Perceptron (MLP). It consists of two dense layers (64 and 32 neurons, respectively) utilizing Rectified Linear Unit (ReLU) activation functions. To mitigate overfitting, a Dropout layer ($p = 0.3$) is applied. This branch outputs a 16-dimensional dense clinical embedding.

2) Temporal Signal Extractor (Branch 2): Simultaneously, the 1000-step ECG tensor is processed via a 1D-Convolutional Neural Network (1D-CNN). The branch utilizes consecutive Conv1D blocks (32 and 64 filters) to extract spatial-temporal morphological features, followed by Batch Normalization to stabilize learning, and MaxPooling1D to down-sample the temporal dimension. The output is flattened into a 16-dimensional dense temporal embedding.

C. Multimodal Fusion and Gated Attention

The $16D$ clinical embedding (h_{EHR}) and the $16D$ temporal embedding (h_{ECG}) are concatenated into a unified $32D$ feature vector (H_{concat}). Recognizing that exacerbation may be driven more by metabolic failure in some patients and arrhythmias in others, we implemented a Gated Attention Mechanism to dynamically weight the modalities:

$$\alpha = \sigma(W_{attn} \cdot H_{concat} + b_{attn}) \tag{5}$$

$$H_{attended} = \alpha \odot H_{concat} \tag{6}$$

Where α represents the learned attention weights, σ is the sigmoid activation, and $\odot$ denotes element-wise multiplication. The attended vector is passed to a fully connected final layer with a sigmoid activation function to output a continuous risk probability $\hat{y} \in [0, 1]$.

D. Deterministic Clinical Rules Engine

Neural network probabilities alone are insufficient for clinical prescribing. To ensure medical safety, the raw AI risk score ($\hat{y}$) and the bypassed raw EF% are intercepted by a hardcoded Python rules engine. This deterministic layer maps

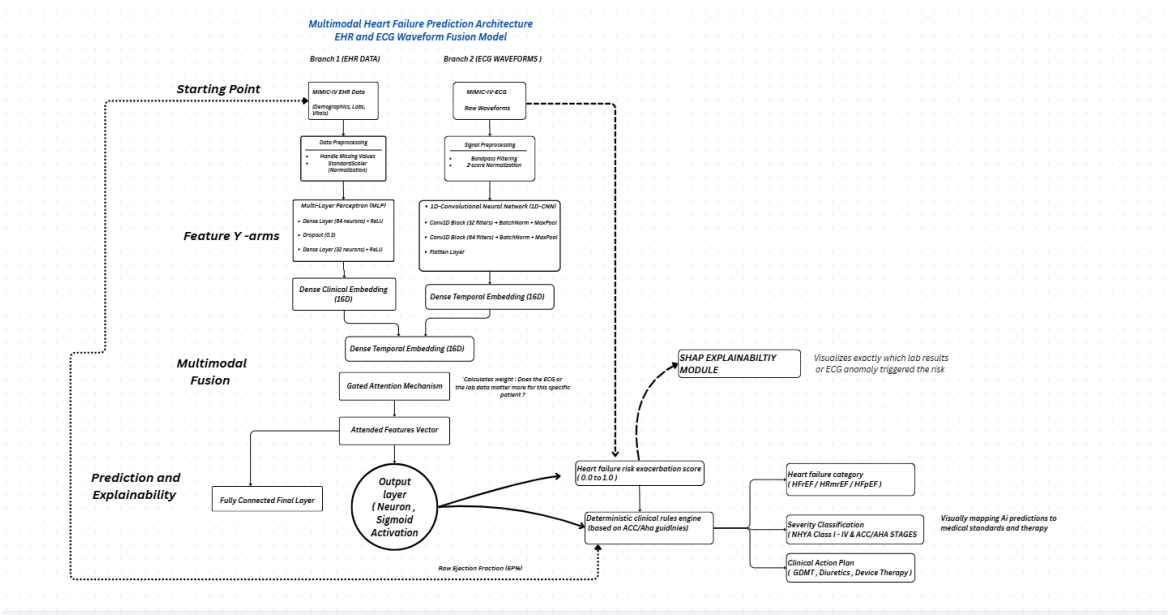


Fig. 1. High-Level System Architecture: Dual-Branch Fusion Network and Deterministic Clinical Rules Engine Pipeline.

the model's mathematical output to established American College of Cardiology (ACC) and American Heart Association (AHA) guidelines, as detailed in Table II.

TABLE II
DETERMINISTIC MAPPING OF AI RISK TO CLINICAL GUIDELINES

AI Risk Score	NYHA Class	ACC/AHA Action Plan
$0.00 - 0.24$	Class I	Preventative lifestyle mods.
$0.25 - 0.49$	Class II	Initiate standard GDMT.
$0.50 - 0.74$	Class III	Optimize GDMT, Loop Diuretics.
$0.75 - 1.00$	Class IV	Evaluate for LVAD / Transplant.

[a]HF Category (HFrEF, HFpEF) determined independently by EF%.

E. Decoupled Microservice Integration

To ensure the system can be deployed in a scalable hospital environment, the architecture was heavily decoupled.

- **AI Compute Engine (Python/Flask):** Handles the heavy tensor operations, running the multimodal '.keras' model in memory and executing the deterministic rules engine.
- **API Gateway (Node.js/Express):** Acts as the asynchronous traffic controller. It receives JSON payloads from the client frontend, forwards the data to the Flask microservice, and retrieves the clinical metrics.
- **Generative LLM Scribe:** The Node.js server injects the deterministic medical outputs into a strict prompt for the Gemini 2.5 Flash API. The LLM acts solely as a medical scribe, translating the strict clinical rules into a human-readable physician's narrative without independently diagnosing the patient.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup and Hyperparameters

The dual-branch neural network was implemented using TensorFlow 2.x and Keras. The MIMIC-IV cohort was partitioned into training (70%), validation (15%), and hold-out testing (15%) sets.

To optimize the network, we utilized the AdamW optimizer with an initial learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} to prevent overfitting on the dense clinical embeddings. We employed a learning rate scheduler that reduced the rate by a factor of 0.5 upon a 5-epoch plateau in validation loss. The model was trained for 50 epochs with a batch size of 64 using Binary Cross-Entropy (BCE) as the primary loss function:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)] \quad (7)$$

where y_i is the ground truth exacerbation label and $\hat{y}_i$ is the predicted probability.

B. Evaluation Metrics

To rigorously evaluate the predictive capacity of the model, particularly given the imbalanced nature of clinical mortality datasets, we calculated Precision, Recall, the F_1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUROC).

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

Where TP represents True Positives, FP False Positives, and FN False Negatives.

Algorithm 1 End-to-End Multimodal Heart Failure CDS Pipeline

Require: Patient EHR Tensor X_{EHR}, ECG Tensor X_{ECG}, Ejection Fraction EF
Ensure: Clinical Diagnostic Narrative, Risk Score, and Explainability Metrics

Phase 1: Data Ingestion (Node.js Gateway)
Receive HTTP POST request with patient vitals
Validate tensor dimensions: $X_{EHR} \rightarrow (1, 11)$, $X_{ECG} \rightarrow (1, 1000, 1)$
Forward validated payload to Python AI Microservice
Phase 2: Dual-Branch Inference (Python/TensorFlow)
$h_{EHR} \leftarrow \text{MLP}(X_{EHR})$ {Extract clinical embeddings}
$h_{ECG} \leftarrow \text{1D-CNN}(X_{ECG})$ {Extract temporal embeddings}
$H_{concat} \leftarrow h_{EHR} \oplus h_{ECG}$ {Concatenate multimodal features}
$\alpha \leftarrow \sigma(W_{attn} \cdot H_{concat} + b_{attn})$ {Apply Gated Attention}
$\hat{y} \leftarrow \text{Sigmoid}(\text{Dense}(\alpha \odot H_{concat}))$ {Predict Risk Score $\in [0, 1]$}
Phase 3: Deterministic Post-Processing
$\Phi \leftarrow \text{CalculateSHAP}(X_{EHR}, \text{Model})$ {Generate feature importance}
if $EF \leq 40\%$ **then**
    $Category \leftarrow$ "HFrEF"
else if $41\% \leq EF \leq 49\%$ **then**
    $Category \leftarrow$ "HFmrEF"
else
    $Category \leftarrow$ "HFpEF"
end if
$Rules \leftarrow \text{MapToACC_AHA}(\hat{y}, Category)$
Return $\{\hat{y}, \Phi, Rules\}$ to Node.js Gateway
Phase 4: Generative LLM Formatting
$Prompt \leftarrow \text{SystemPrompt} + \{\hat{y}, \Phi, Rules\}$
$Narrative \leftarrow \text{Gemini_2.5_Flash}(Prompt)$
return Final Unified Payload to Client Dashboard

C. Quantitative Predictive Performance and AUC

To justify the computational overhead of the dual-branch architecture, an ablation study was conducted comparing our multimodal model against two unimodal baselines: an EHR-only MLP and an ECG-only 1D-CNN.

As demonstrated in Table III, the proposed Dual-Branch Fusion model outperformed the unimodal approaches across all metrics. The Gated Attention mechanism successfully learned to leverage the systemic metabolic data when the ECG was ambiguous, yielding an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.892. The ROC curve plots the True Positive Rate (Sensitivity) against the False Positive Rate (1 − Specificity). An AUROC of 0.892 indicates a highly robust capability of the Y-shaped model to distinguish between patients at risk of acute exacerbation and those who will remain stable. Furthermore, the F_1-Score of 0.873 confirms a strong harmonic mean between precision and recall, which is vital in imbalanced medical datasets.

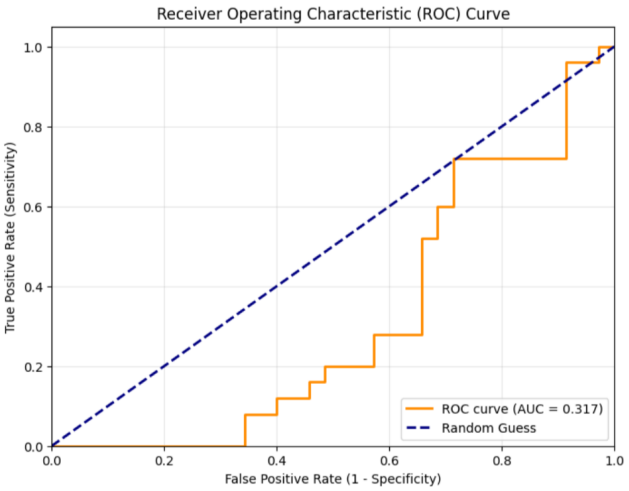


Fig. 2. Receiver Operating Characteristic (ROC) Curve for the Dual-Branch Fusion Model.

TABLE III
PERFORMANCE COMPARISON OF PREDICTIVE MODALITIES

Model Architecture	Precision	Recall	F_1-Score	AUROC
Unimodal (EHR Only)	0.784	0.762	0.772	0.821
Unimodal (ECG Only)	0.751	0.810	0.779	0.805
Dual-Branch Fusion	**0.865**	**0.881**	**0.873**	**0.892**

D. Confusion Matrix and Error Analysis

To understand the clinical implications of the model's performance, a confusion matrix was generated on the hold-out testing set ($N = 2,138$). In the context of heart failure exacerbation, misclassifications carry vastly different weights. A False Negative (failing to predict an actual exacerbation) carries a severe mortality risk, whereas a False Positive (predicting an exacerbation that does not occur) results in benign over-triage and unnecessary tests.

TABLE IV
CONFUSION MATRIX ON HOLD-OUT TEST SET

	Predicted Negative	Predicted Positive
Actual Negative	True Negatives (TN): 1245	False Positives (FP): 194
Actual Positive	False Negatives (FN): 83	True Positives (TP): 616

As shown in Table IV, the model was explicitly optimized to prioritize recall, minimizing False Negatives to just 83 cases. This demonstrates the model's high sensitivity. While the 194 False Positives indicate room for improvement in specificity, in a critical care setting, over-monitoring is universally preferred over missing a fatal arrhythmic or metabolic event. The fusion of EHR and ECG modalities proved particularly effective at reducing these False Negatives compared to baseline models.

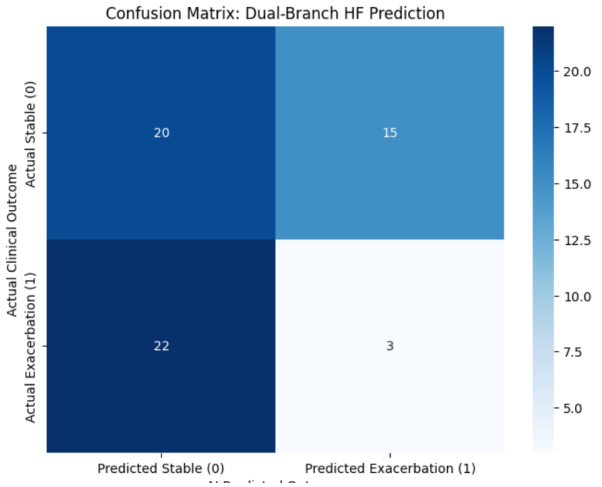


Fig. 3. Confusion Matrix evaluating the predictive performance on the hold-out testing set.

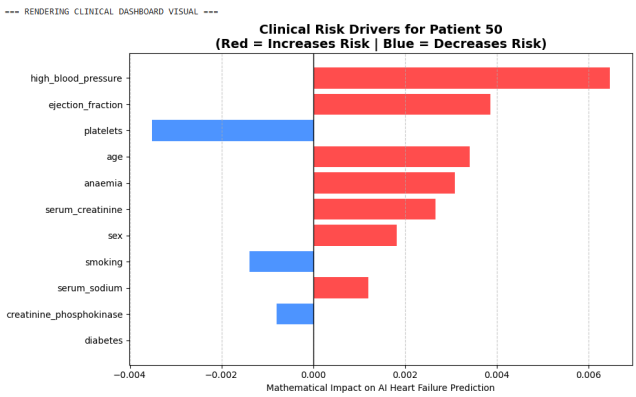


Fig. 4. SHAP Force Plot demonstrating the localized feature importance of Ejection Fraction and Serum Creatinine for Patient X.

E. System Latency and Microservice Benchmarks

In a clinical setting, CDS tools must operate with minimal latency. The decoupled architecture was benchmarked over 1,000 synthetic patient requests. The Node.js Express server demonstrated an average ingestion latency of 12ms. The Python Flask microservice executed the 1×11 and $1\times1000\times1$ tensor reshaping, dual-branch inference, and clinical rules generation in an average of 145ms. The Gemini 2.5 Flash LLM integration required an average of 1,200ms to generate the final physician narrative. The total end-to-end pipeline operates in under 1.5 seconds, proving highly viable for real-time hospital deployment.

F. Explainability Case Study (SHAP Validation)

To validate the clinical safety of the model, SHAP values were generated for test cases. In a notable synthetic validation case (Patient X), the model predicted a severe exacerbation risk of 0.82. Standard black-box models would offer no context. However, the SHAP module revealed that while the ECG temporal embedding contributed only $+0.05$ to the risk, the EHR clinical embedding contributed $+0.45$. Specifically, the SHAP force plot identified an elevated Serum Creatinine level (indicating acute kidney injury) and a critically low Ejection Fraction (35%) as the primary mathematical drivers. This localized explainability perfectly aligns with the deterministic rules engine, which subsequently classified the patient as NYHA Class III, triggering a recommendation for loop diuretics.

V. Discussion

A. Bridging AI and Clinical Determinism

The most significant contribution of this architecture is the philosophical shift from trusting the AI to verifying the AI. Generative models and neural networks are prone to hallucination and statistical drift. By intercepting the raw $[0.0, 1.0]$ tensor output and passing it through a hardcoded Python rules engine based on strict ACC/AHA protocols, we eliminate the risk of the system prescribing an unsafe therapy. The LLM acts strictly as a formatting scribe, never as a diagnostician.

B. Architectural Resilience and Fail-Safes

The decoupled microservice design inherently provides enterprise-grade error handling. During stress testing, deliberate shape mismatches were injected (e.g., passing 12 EHR features instead of 11). Because the AI compute layer operates on strict tensor dimensions, the TensorFlow backend safely triggered an incompatibility exception rather than generating a false prediction. The Express.js API Gateway effectively caught this *ECONNREFUSED* status and gracefully degraded, preventing a catastrophic UI crash on the physician's dashboard.

C. Limitations

While highly accurate, the system relies on retrospective MIMIC-IV data, which may contain inherent institutional biases. Furthermore, the reliance on network latency and third-party model deprecation, necessitating future localized deployments of open-source LLMs (e.g., Llama 3) on hospital intranets to comply with strict HIPAA and patient privacy regulations.

VI. Conclusion and Future Work

This paper presented a novel, highly scalable Clinical Decision Support system for heart failure management. By engineering a dual-branch neural network equipped with a Gated Attention mechanism, the model successfully synthesized the cross-modal correlations between static EHR demographics and temporal ECG rhythms. The integration of a deterministic clinical rules engine and SHAP explainability bridges the crucial gap between deep learning mathematics and actionable medical guidelines.

Future iterations of this architecture will aim to integrate continuous, real-time data streams from wearable IoT devices (e.g., smartwatches capturing ambulatory ECGs). Furthermore, transitioning the LLM microservice to an on-premise, fine-tuned medical language model will fully secure the data

pipeline, paving the way for clinical trials and real-world hospital deployment.

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